Unsymmetric spectral elements under non-polynomial displacement fields: distortion sensitivity and convergence for the thick cylinder

USEM-MATLAB study `studies/thick_cylinder`

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Abstract

The unsymmetric (parametric test / metric trial) spectral element of the USEM code base is studied for the thick cylinder of Rajendran [5, Sec. 5.9] with $p = 2, \ldots, 8$ on gmsh meshes whose distortion is kept fixed under refinement ($\alpha = 0°$–$90°$, including folded elements), complemented by manufactured polynomial and analytic non-polynomial solutions on the same meshes; the conventional spectral element serves as reference (1008 solutions, verified against the code base). *Findings.* (i) A degree-5 polynomial field is reproduced up to round-off for $p \geq 5$ on all meshes without folds, but the round-off level grows with distortion and p and destroys the exactness on folded meshes. (ii) For non-polynomial fields distortion immunity is lost, but for moderate distortion ($\alpha = 15°$, all $p \leq 6$; $\alpha = 30°$, $p \leq 3$) the element remains distortion tolerant (energy error at most 1.84 times that on the undistorted mesh) and converges with the optimal order p, whereas the conventional element drops to $\lfloor p/2 \rfloor$. (iii) For stronger distortion or higher p both distortion tolerance and optimal convergence are lost; convergence becomes erratic and the element is less accurate than the conventional one, independently of the sign of $\det \boldsymbol{J}$. The loss is caused by the ill-conditioned metric interpolation (Lebesgue constants up to 10^{12}), which degrades the stability of the Petrov–Galerkin operator. The practical benefit is therefore confined to moderately distorted, non-folding meshes; the element is not a reliable remedy for severely distorted or folded meshes at high order.

1 Objective

Unsymmetric (Petrov–Galerkin) elements with parametric test and metric trial functions reproduce every displacement field of total degree $\leq p$ exactly on arbitrarily distorted meshes, including meshes with $\det \boldsymbol{J} \leq 0$ (*distortion immunity*) [5, 6]. Displacement fields of practical problems are, however, not polynomial. Rajendran [5] observed for the thick cylinder (Section 5.9 of that paper, 8-node element) that the energy-norm convergence rate decreases from about 2 to 1.85, 1.42 and 1.21 as the distortion angle increases from 0° to 30°, 45° and 60°, and concluded that conditions C1–C3 do not guarantee optimal convergence for non-polynomial fields. This study revisits the question for the unsymmetric spectral elements of the code base, $p = 2, \ldots, 8$:

1. Is distortion resistance lost for non-polynomial fields?
2. If so, are optimal convergence rates nevertheless retained?
3. If both are lost, what is the practical benefit of unsymmetric elements?

The thick cylinder is complemented by manufactured solutions on the same geometry and meshes: a polynomial field (degree 5), for which the element is immune for $p \geq 5$, and an analytic non-polynomial field. All statements below are based on the computed data in `results/` (reproducible with the scripts listed in Section 7); the conventional spectral element on the same meshes serves as reference.

2 Formulation and set-up

2.1 Elements

Both elements use the same geometry, the same nodes and the same test functions: Lagrange polynomials $N_i(\boldsymbol{\xi}) \in Q_p$ on the $(p+1)^2$ Gauss–Lobatto–Legendre (GLL) nodes of the reference square, mapped isoparametrically, $\boldsymbol{x}(\boldsymbol{\xi}) = \sum_i N_i(\boldsymbol{\xi})\, \boldsymbol{x}_i$. They differ in the trial functions.

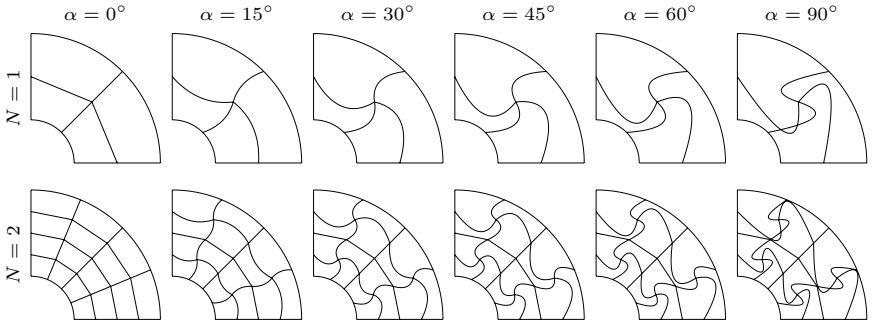


Figure 1: Meshes generated with gmsh for $N = 1$ (2×2 elements) and $N = 2$ (4×4 elements); the internal mid-side nodes of every super-element are rotated by α. Meshes for $N = 4, 8$ repeat the pattern on N^2 super-elements.

- PM (unsymmetric spectral element of the code base, `Ke_2D.m`): the trial functions are *metric* functions $M_i^e(\boldsymbol{x}) \in Q_p(\hat{x}, \hat{y})$, defined element by element through $M_i^e(\boldsymbol{x}_j) = \delta_{ij}$ in the corotational skew coordinates $(\hat{x}, \hat{y})^{\mathsf{T}} = \boldsymbol{J}_0^{-\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}_0)$ ($\boldsymbol{J}_0$, $\boldsymbol{x}_0$: Jacobian and image of the element centre) [7]. As $(\hat{x}, \hat{y})$ is affine in $\boldsymbol{x}$, the span of $\{M_i^e\}$ contains $P_p(\boldsymbol{x})$ (all polynomials of total degree $\leq p$ in Cartesian coordinates) for *every* element shape. With $\boldsymbol{B}_L^* = \boldsymbol{B}_L \det \boldsymbol{J}$ (built from $\operatorname{adj} \boldsymbol{J}\, \nabla_{\boldsymbol{\xi}} N_i$),

$$\boldsymbol{K}^e = \int_{-1}^{1}\int_{-1}^{1} (\boldsymbol{B}_L^*)^{\mathsf{T}} \boldsymbol{D}\, \boldsymbol{B}_R \, \mathrm{d}\xi \, \mathrm{d}\eta, \tag{1}$$

  i.e. the Jacobian determinant cancels and the integrand is a polynomial in $\boldsymbol{\xi}$; this is the Jacobian-unlimited formulation PM-QUAD8* of Rajendran [5] generalised to degree p (conditions C1–C3 of that paper). The trial space is non-conforming (the M_i^e are not continuous across element edges).

- PP (conventional spectral element, reference): parametric functions as test and trial functions, $\boldsymbol{K}^e = \int \boldsymbol{B}_L^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_L \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta$ (`tests/helpers/usemKeSymmetric.m`).

The load vectors are identical for both elements (parametric test functions), $\boldsymbol{f}^e = \int N_i \, \boldsymbol{b} \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta + \int_{\Gamma_N} N_i \, \bar{\boldsymbol{t}} \, \mathrm{d}s$. The stiffness matrices use the default rule of `Ke_2D.m`, $2p \times 2p$ Gauss points; loads and error integrals use $(3p+2)^2$ points. The element matrices of the study (`cyl_Ke_PM.m`, `cyl_Ke_PP.m`) are vectorised versions of `Ke_2D.m` and `usemKeSymmetric.m`; they agree with them to round-off (Table 1).

2.2 Model and mesh (gmsh)

Thick cylinder under internal pressure (Section 5.9 of [5]): quarter annulus $a = 1 \leq r \leq b = 3$, $E = 1000$, $\nu = 0.25$, plane stress, internal pressure $p_i = 1000$, symmetry conditions $u_x = 0$ on $x = 0$ and $u_y = 0$ on $y = 0$. The geometry (exact circular arcs, built-in kernel) and the structured mesh are generated with gmsh [3] (`mesh/gen_cylinder_meshes.py`): a transfinite mesh of $2N \times 2N$ eight-node quadrilaterals, i.e. $N \times N$ super-elements of 2×2 elements ($N = 1, 2, 4, 8$; element size $h \propto 1/N$). Mesh distortion follows mesh sequence B of [5, Fig. 19]: in every super-element the mid-side nodes of the four internal edges are rotated counter-clockwise about the centre node by the angle $\alpha \in \{0°, 15°, 30°, 45°, 60°, 90°\}$ (Fig. 1). The pattern is repeated on every super-element, so the *relative* distortion does not decrease under refinement: the element maps are not asymptotically affine. For $\alpha \geq 60°$ (and $\alpha = 45°$ for $N \geq 2$), $\det \boldsymbol{J}$ changes sign inside elements (folded elements, Table 2). Each mesh is written as an ABAQUS input file (corner nodes first, node and element sets, pressure `*Dsload`) that also runs with `code/Start_Sim.m`.

The 8-node mesh is lifted to degree p by `pSpectralElement_NodesTopology.m`. Element edges on $r = a$ and $r = b$ are then replaced by the exact arcs (Gordon–Hall blending [4]); interior edges remain the parabolas of the 8-node mesh. The degree-p geometry thus interpolates the circle at the GLL points; the signed area converges to 2π spectrally ($|A_h - 2\pi| < 10^{-14}$ for $p = 8$) and is independent of α because folded parts cancel (Table 1).

2.3 Problems

1. *Lamé* (non-polynomial, rational): $u_r = \frac{a^2 p_i \, r}{E(b^2 - a^2)} \left[(1-\nu) + (1+\nu)\frac{b^2}{r^2}\right]$, $u_\theta = 0$ [5, Eq. (25)]; symmetry conditions on the straight edges, tractions on the arcs.

2. *Manufactured polynomial* field of total degree 5 ($X = x/3$, $Y = y/3$):
   $u_x = \frac{1}{10}(1 + X + 2Y + XY - 3X^2Y + X^3 + 2X^4Y - X^2Y^3 + X^5 - 2Y^5)$,
   $u_y = \frac{1}{10}(1 - X + Y^2 + 2X^2Y - X^3 + XY^3 - 3X^3Y^2 + Y^5 + 2XY^4 - X^5)$.

3. *Manufactured smooth non-polynomial* (entire) field: $u_x = \frac{1}{10}\big(\sin 3X \cos 2Y + e^{XY}\big)$, $u_y = \frac{1}{10}\big(\cos 2X \sin 3Y + Y e^{-X}\big)$.

For the manufactured solutions (MMS) both displacement components are prescribed on $x = 0$ and $y = 0$ (nodal values) and the tractions $\boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n}_h$ on the arcs. Strains, stresses and the body force $\boldsymbol{b} = -\operatorname{div}\boldsymbol{\sigma}(\boldsymbol{u})$ are derived symbolically with the operators of the MMS module of the code base (`MMS_2D/differentialOperators`, as in `examples_Elasticity.m`); for the Lamé field this yields $\boldsymbol{b} \equiv \boldsymbol{0}$. On the arcs $\boldsymbol{n}_h$ is the normal of the discrete boundary, so that the discrete problem on Ω_h has the exact field as solution whenever it lies in the trial space (with the exact normal, the polynomial patch test fails at the level of the geometry error, e.g. $e_E \approx 3 \cdot 10^{-6}$ for $p = 5$, $N = 1$).

2.4 Error measures

With the strain field of the respective trial functions (metric for PM, parametric for PP),

$$e_E^2 = \frac{\sum_e \int_{\hat{\Omega}} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_h^e)^{\mathsf{T}} \boldsymbol{D} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_h^e) \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta}{\int_\Omega \boldsymbol{\varepsilon}^{\mathsf{T}} \boldsymbol{D} \boldsymbol{\varepsilon} \, \mathrm{d}\Omega}, \qquad e_{L^2}^2 = \frac{\sum_e \int_{\hat{\Omega}} |\boldsymbol{u} - \boldsymbol{u}_h^e|^2 \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta}{\int_\Omega |\boldsymbol{u}|^2 \, \mathrm{d}\Omega}, \tag{2}$$

and the nodal error $e_n = \max_i |\boldsymbol{u}_h(\boldsymbol{x}_i) - \boldsymbol{u}(\boldsymbol{x}_i)| / \max_i |\boldsymbol{u}(\boldsymbol{x}_i)|$. The signed $\det \boldsymbol{J}$ is used, consistent with (1): on folded elements each element contributes with the degree of its map. For PP on folded meshes the parametric strain is singular where $\det \boldsymbol{J} = 0$, so e_E is not meaningful there; e_n is defined for all cases. Observed rates are $r = \log_2(e_N / e_{2N})$. For smooth solutions and shape-regular, asymptotically affine meshes, optimal rates are p (e_E) and $p + 1$ (e_{L^2}) [2].

3 Verification

Table 1 summarises the cross-checks of the study routines against the code base (`matlab/cyl_verify.m`, full output in `results/verification.txt`). The study was run with GNU Octave 8.4 (MATLAB is not available in the build environment); three Octave stand-ins (`matlab/octave_compat`) replace MATLAB functions used by the code base (`contains`, `decomposition`, the third output of `unique(...,'stable')`). Table 2 lists the number of folded elements.

Table 1: Verification against the code base (relative differences).

check	result
element stiffness: `cyl_Ke_PM` vs `Ke_2D`, `cyl_Ke_PP` vs `usemKeSymmetric` ($\alpha = 0°, 60°, 90°$; $p = 2, 5, 8$)	$\le 5.4 \cdot 10^{-15}$ (PM), $\le 5.7 \cdot 10^{-12}$ (PP, folded elements)
global stiffness vs `AssemblyFEM` ($N = 2$; $\alpha = 0°, 60°$; $p = 2, 4$)	$\le 1.1 \cdot 10^{-15}$
pressure load of the ABAQUS file (`NeumannBCs/CreatePress`) vs $\int N \boldsymbol{\sigma}\boldsymbol{n}_h \, \mathrm{d}s$ (Lamé, $N = 1$, $p = 2, \ldots, 8$)	$2 \cdot 10^{-3}$ ($p = 2$) $\to 1.7 \cdot 10^{-11}$ ($p = 8$): geometry error of the arc; resultant $1000 = p_i a$ in both
static solution with the code-base pipeline (`AssemblyFEM`, `NeumannBCs`, BCs of the ABAQUS file) vs study routines	$\le 2.3 \cdot 10^{-11}$ ($\alpha = 0°$), $\le 4.8 \cdot 10^{-7}$ ($\alpha = 60°$, round-off amplified by the conditioning)
signed area -2π ($p = 8$, all meshes)	$\le 1.9 \cdot 10^{-14}$
exact fields (Lamé): $\boldsymbol{b} = \boldsymbol{0}$, $\sigma_r(a) = -p_i$, $\sigma_\theta(a) = 1250$, $\sigma_r(b) = 0$, $u_r(a) = 1.5$	exact
polynomial patch test, PM, $p \ge 5$, $\alpha \le 30°$ (Table 3)	$e_n \le 8 \cdot 10^{-9}$

Table 2: Folded elements ($\det \boldsymbol{J} \le 0$ somewhere in the element) / number of elements.

	$N = 1$	$N = 2$	$N = 4$	$N = 8$
$\alpha = 0°$	0/4	0/16	0/64	0/256
$\alpha = 15°$	0/4	0/16	0/64	0/256
$\alpha = 30°$	0/4	0/16	0/64	0/256
$\alpha = 45°$	0/4	2/16	8/64	32/256
$\alpha = 60°$	2/4	6/16	32/64	112/256
$\alpha = 90°$	4/4	16/16	64/64	256/256

Table 3: Polynomial field (degree 5), PM: relative nodal error e_n for $p \geq 5$ (exact in exact arithmetic).

	$N = 1$				$N = 8$			
	$p = 5$	$p = 6$	$p = 7$	$p = 8$	$p = 5$	$p = 6$	$p = 7$	$p = 8$
$\alpha = 0°$	2.3e-14	8.7e-14	4.0e-12	3.3e-11	1.4e-13	7.7e-14	2.8e-13	4.6e-13
$\alpha = 15°$	1.1e-11	4.9e-13	9.6e-12	3.7e-11	7.4e-14	2.2e-13	1.6e-11	1.6e-11
$\alpha = 30°$	4.4e-12	1.7e-12	3.5e-11	1.8e-9	5.0e-11	4.3e-11	2.5e-9	7.7e-9
$\alpha = 45°$	5.3e-12	3.2e-9	9.8e-9	3.1e-6	2.1e-8	2.0e-7	1.7e-5	1.1e-4
$\alpha = 60°$	7.9e-11	6.1e-9	5.9e-8	1.5e-6	7.4e-1	8.5e1	3.5e6	1.7e9
$\alpha = 90°$	3.4e-7	2.2e-5	3.5e-3	4.0e0	5.9e-4	3.2e0	1.3e2	1.9e3

4 Results

The complete data (6 angles × 4 meshes × 7 degrees × 2 elements × 3 problems = 1008 solutions) are in `results/raw/study_a*.csv`. Rates are given for the finest pair $N = 4 \to 8$; "–" marks errors at the round-off level (for PM and $p \geq 5$: less than ten times the error of the polynomial field on the same mesh, which is the empirical round-off level, Section 4.2); † marks angles with folded elements on the meshes involved (Table 2).

4.1 Reproduction of Section 5.9 for $p = 2$

For $p = 2$ and the meshes $N = 1, 2, 4$ of the paper, the least-squares slopes of e_E (Lamé) are 1.90, 1.95, 1.14, 1.57 for PM and 1.68, 1.11, 1.07, 0.69 for PP at $\alpha = 0°, 30°, 45°, 60°$ (Rajendran [5, Fig. 20]: PM-QUAD8* 2.1, 1.85, 1.42, 1.21; QUAD8 1.9, 0.95, 0.94, 0.98). The trends agree: the conventional element drops to first order on distorted meshes, the unsymmetric element stays close to second order for $\alpha \leq 30°$ and becomes erratic beyond. Exact agreement is not expected, since the present $p = 2$ element is the 9-node Lagrange element with Q_2 metric functions in skew coordinates, not the 8-node serendipity element of the paper. On the finer pair $N = 4 \to 8$, PM ($p = 2$) recovers rate ≈ 2 for all $\alpha \leq 45°$ (Tables 5 and 7).

4.2 Polynomial field: distortion immunity

For $p \geq 5$ the degree-5 field lies in the PM trial space on every element. Table 3 shows the nodal error: on meshes with $\det \boldsymbol{J} > 0$ and moderate distortion ($\alpha \leq 30°$) it is 10^{-14}–10^{-8}, i.e. the field is reproduced up to round-off on every mesh, as predicted. The round-off level, however, grows with α and p and reaches 10^{-4} at $\alpha = 45°$, $p = 8$. On the folded meshes ($\alpha \geq 60°$) the computed solutions for $p \geq 5$ are useless (nodal errors up to 10^9), although the method is exact in exact arithmetic. A dedicated test (`results/conditioning.txt`) integrates the PM stiffness exactly ($n_q \geq p^2 + p/2$ points, Section 4.6) and evaluates the residual $\boldsymbol{K}\boldsymbol{u}_I - \boldsymbol{F}$ of the exact nodal values, which vanishes in exact arithmetic: it is $4 \cdot 10^{-14}$ at $\alpha = 0°$, $p = 5$, but $2 \cdot 10^{-2}$ ($\alpha = 60°$, $N = 2$, $p = 8$) and 49 ($\alpha = 90°$, $N = 1$, $p = 8$), with $\sigma_{\min}/\sigma_{\max}(\boldsymbol{K}_{\mathsf{PM}}) = 10^{-15}$ against $3 \cdot 10^{-6}$ for PP. Distortion immunity therefore holds in exact arithmetic but not in floating point once the metric interpolation becomes ill-conditioned (Lebesgue constants up to $5 \cdot 10^{12}$, Table 9). For $p < 5$ the polynomial field behaves like a non-polynomial one (rates $2, 3, 4$ for $\alpha \leq 15°$, Fig. 7).

4.3 Non-polynomial fields: convergence

Figures 2 and 3 show the nodal error, Fig. 4 the energy error for the meshes without folded elements; the rates are collected in Tables 4 to 7. Three regimes can be distinguished.

1. $\alpha = 0°$ (element maps asymptotically affine): both elements converge optimally, $e_E = O(h^p)$ (smooth field: 2.00, 2.99, 4.00, 4.99, 6.00 for PM; Lamé: 1.97–6.62, approaching p from below as the r^{-2} term is resolved), with almost identical errors.

2. $\alpha = 15°$: PM keeps the optimal rates for all p (smooth: 2.00, 2.99, 3.99, 4.99, 5.98; Lamé: 1.97, 2.91, 3.90, 4.81, 5.83) and for $p \leq 6$ its error exceeds that on the undistorted mesh by only 5–12 % (smooth) and 5–28 % (Lamé) in e_E, and by a factor 1.5–4.7 in e_n (Table 8). For $p \geq 7$ the PM error reaches a round-off floor of about 10^{-10}, two to three orders above the floor on the undistorted mesh. PP loses half of the order: its energy rates are 1.08, 1.23, 2.21, 2.03, 2.94, 3.39, 4.05 (smooth) and, for the polynomial field, 0.98, 0.95, 1.95, 1.94, 2.94, 2.95, 3.95, i.e. $\lfloor p/2 \rfloor$. This is the approximation property of the parametric space on maps that are not asymptotically affine: the mid-side node offset is $O(h)$, so $\boldsymbol{x}(\boldsymbol{\xi})$ is quadratic with $O(1)$ relative non-affinity, and $Q_p \circ \boldsymbol{x}^{-1}$ contains $P_k(\boldsymbol{x})$ only for $2k \leq p$ (cf. [1, 2]). At $N = 8$, $p = 6$ the PM error is 390 times smaller than the PP error (smooth, e_E).

3. $\alpha \geq 30°$: PM is optimal only for $p \leq 3$ at 30° (and for $p = 2$ at 45° for the Lamé field, rate 1.94; smooth field 1.11); for higher p the convergence becomes erratic and even divergent (e.g. smooth, $\alpha = 30°$, $p = 4$: $e_E = 2.2 \cdot 10^{-4}$ at $N = 4$ and $4.3 \cdot 10^{-4}$ at $N = 8$; $\alpha = 45°$, $p = 8$: $e_E = 3 \cdot 10^{-3}$ at $N = 8$). On the folded meshes ($\alpha \geq 60°$) PM fails for $p \geq 5$ (Section 4.2). PP converges monotonically at the reduced rate $\approx \lfloor p/2 \rfloor$ in e_E (nodal error: $\approx \lfloor p/2 \rfloor + 1$) for all $\alpha \leq 45°$ and all p (energy norm; the nodal error with one exception, $p = 5$ at 45°). On the folded meshes $\alpha \geq 60°$ its convergence is irregular for $p \leq 3$, but the nodal error still decreases by 1.5–3 orders of magnitude between $N = 1$ and $N = 8$ for $p \geq 4$ (Fig. 2).

Under p-refinement on a fixed mesh (Fig. 6), PM converges exponentially for $\alpha \leq 15°$ down to its round-off floor (reached at $p \approx 6$ for $N = 8$), like PP on the undistorted mesh. At $\alpha = 30°$ and 45° the PM error is non-monotone in p (smooth field, $N = 8$, $\alpha = 30°$: e_n jumps from $2.4 \cdot 10^{-6}$ at $p = 3$ to $2.8 \cdot 10^{-5}$ at $p = 4$ and grows from $9.4 \cdot 10^{-11}$ at $p = 6$ to $8.0 \cdot 10^{-9}$ at $p = 8$), while the PP error decreases monotonically, although more slowly than on the undistorted mesh.

4.4 Distortion sensitivity

Table 8 gives the error amplification $e_n(\alpha)/e_n(0°)$ on the finest mesh, Fig. 5 the nodal error versus α. For a non-polynomial field neither element is distortion immune: the PM error depends on α. For $\alpha = 15°$ and $p \leq 6$ the PM amplification is 1.5–2.7 (smooth) and 3.2–4.7 (Lamé), against 4.8–99 and 61–2900 for PP. At 30° both elements are amplified by similar factors (50–$2 \cdot 10^3$ for $p = 3$–6; PM smaller for some p, larger for others). For $\alpha \geq 45°$ and $p \geq 5$ the PM amplification exceeds the PP amplification by one to ten orders of magnitude. (For $p \geq 7$ the reference errors at $\alpha = 0°$ are at round-off level, so these columns underestimate the sensitivity of PP and overstate that of PM.)

4.5 Mechanism

The diagnostics in `results/` identify where the loss originates.

1. *Not the Jacobian, not quadrature.* The loss occurs on meshes with $\det \boldsymbol{J} > 0$ ($\alpha = 30°$) and persists with exact integration (Sections 4.2 and 4.6).

2. *Not consistency.* The residual of the exact nodal values, $\boldsymbol{r} = \boldsymbol{K}\boldsymbol{u}_I - \boldsymbol{F}$, is *smaller* for PM than for PP in all eleven cases tested (`residual_smooth.txt`; e.g. $\alpha = 30°$, $N = 8$, $p = 4$: $3.0 \cdot 10^{-5}$ versus $3.7 \cdot 10^{-4}$), whereas the PM nodal error is 9 times larger ($2.8 \cdot 10^{-5}$ versus $3.0 \cdot 10^{-6}$). The ratio of error to residual of PM grows under refinement (0.05 at $N = 4$, 0.9 at $N = 8$).

3. *Stability of the unsymmetric operator.* With $\boldsymbol{u}_h - \boldsymbol{u}_I = -\boldsymbol{K}^{-1}\boldsymbol{r} = -\sum_k \boldsymbol{v}_k(\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{r})/\sigma_k$ (singular triplets of $\boldsymbol{K}$; `spectrum_smooth.txt`, smooth field), the error of PP and of PM for $\alpha \leq 15°$ is spread over the spectrum (largest single contribution 2–8 % of $\|\boldsymbol{u}_h - \boldsymbol{u}_I\|^2$, from mid-spectrum modes). For $\alpha \geq 30°$, $p = 4$, PM has singular values below the smallest singular value of PP (down to $0.17\,\sigma_1^{\mathrm{PP}}$), and in four of the five cases tested a single one of the lowest six triplets carries 21–68 % of the error (the fifth, $\alpha = 30°$, $N = 4$, $p = 6$: triplet $k = 28$, 9 %). The residual of the non-conforming Petrov–Galerkin scheme is not orthogonal to these low modes, unlike the Galerkin residual of PP, so they are excited and amplified.

4. *Ill-conditioned metric interpolation.* The Lebesgue constant $\Lambda_M = \max_{\boldsymbol{x} \in \Omega^e} \sum_i |M_i^e(\boldsymbol{x})|$ of the metric functions (basis independent) equals the GLL value on affine elements, but grows by orders of magnitude with α and p (Table 9): $\Lambda_M \leq 163$ for $\alpha = 15°$, $p \leq 6$, but $1.1 \cdot 10^3$ ($\alpha = 30°$, $p = 4$), $9 \cdot 10^5$ (45°, $p = 5$) and $5 \cdot 10^{12}$ (60°, $p = 8$). The element condition numbers of PM exceed those of PP by similar factors (Table 10). The images of the GLL nodes of strongly non-affine elements come close to configurations that are not unisolvent for Q_p in skew coordinates ("interpolation failure" in [5]); the metric trial functions then have large gradients, the bilinear form loses its stability constant, and round-off is amplified (Section 4.2). All cases of the benign regime of Section 4.3 have $\Lambda_M \leq 2.5 \cdot 10^2$, all degraded cases $\Lambda_M \geq 10^3$; the converse does not hold strictly ($\alpha = 30°$, $p = 6$: $\Lambda_M = 1.1 \cdot 10^4$, yet PM is 500 times more accurate than PP at $N = 8$), so Λ_M is a necessary rather than a sufficient indicator. The maximum over the mesh does not decrease under refinement because the distortion is self-similar; it varies with N because the element shapes near the curved boundaries change.

4.6 Quadrature

The PM integrand (1) is a polynomial of degree $(2p - 1)(q + 1)$ per direction for a geometry of degree q; exact integration needs $n_q \geq p^2 + p/2$ points on the elements with exact arcs ($q = p$) and $n_q \geq 3p - 1$ on

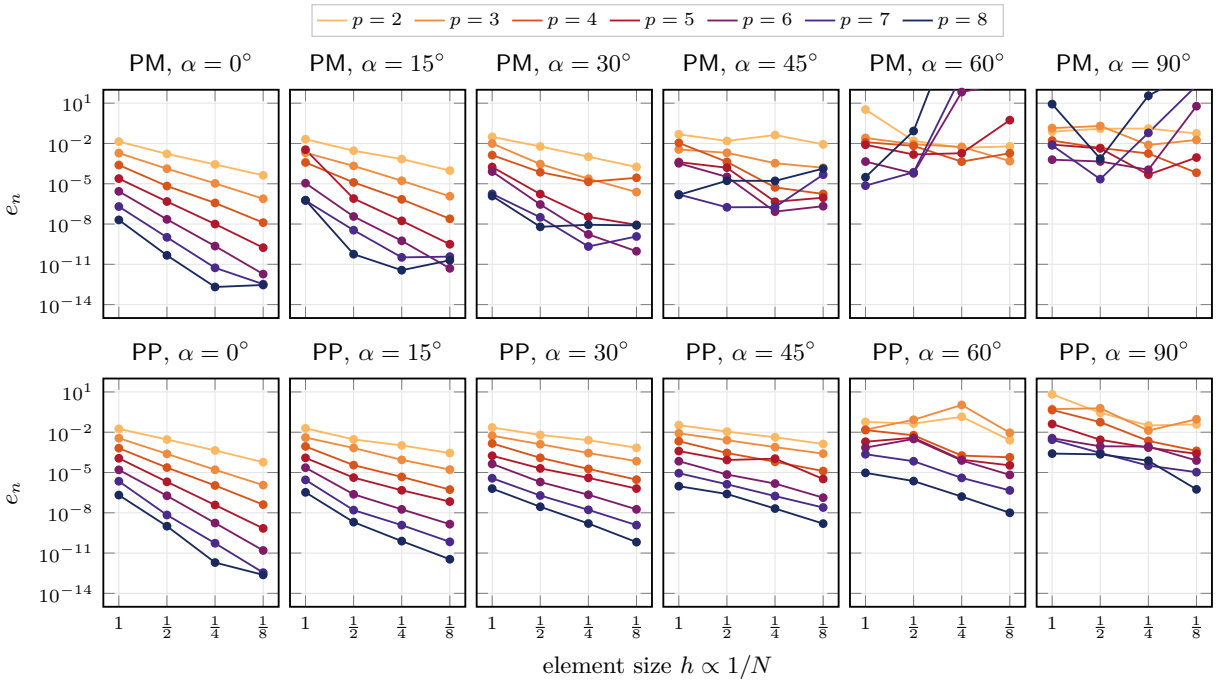


Figure 2: Smooth manufactured field: nodal error e_n versus element size for all distortion angles, PM (top) and PP (bottom).

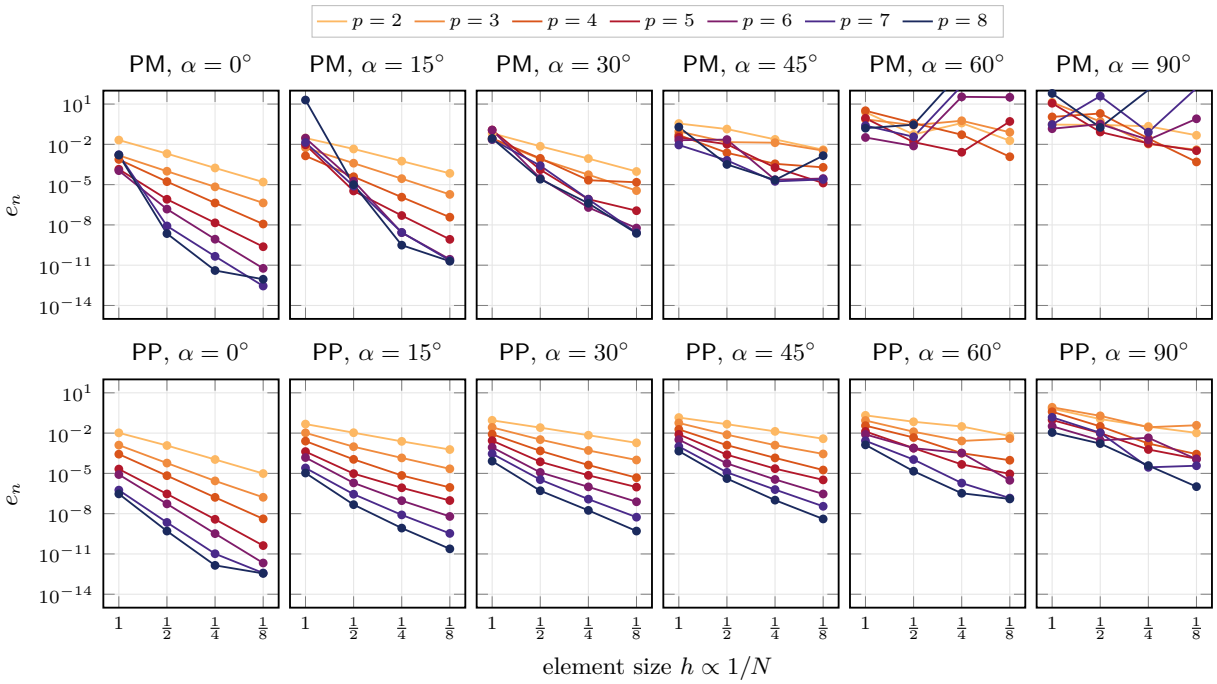


Figure 3: Thick cylinder (Lamé): nodal error e_n versus element size, PM (top) and PP (bottom).

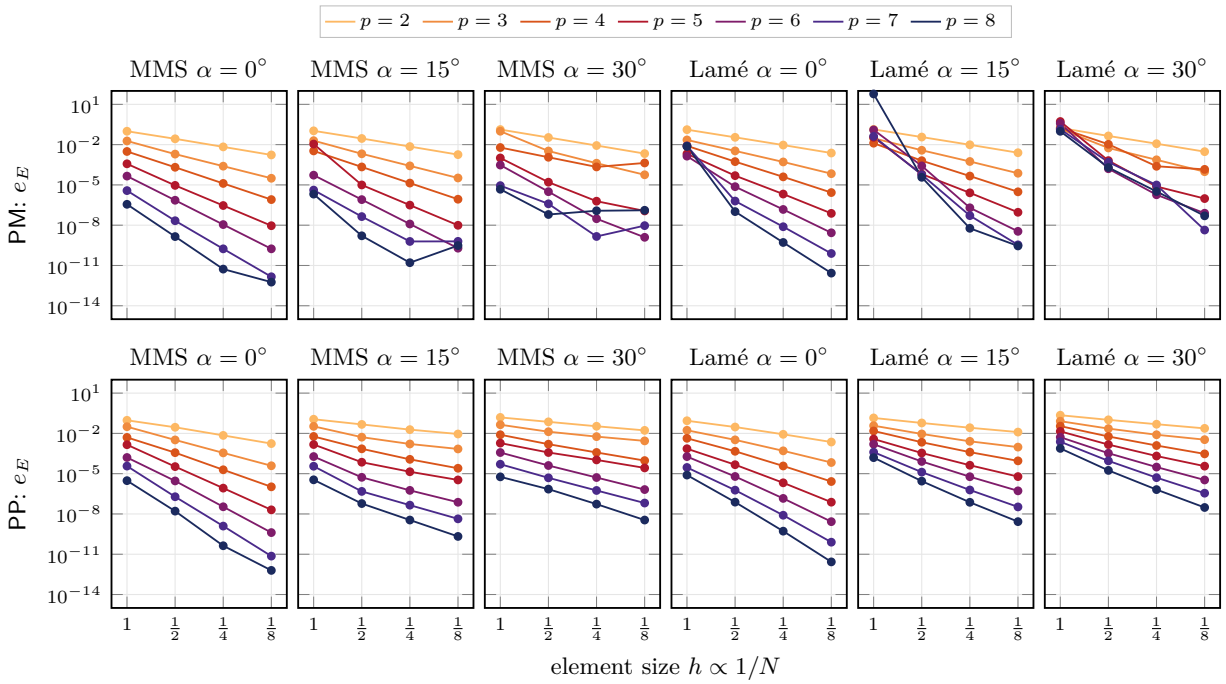


Figure 4: Energy error e_E on the meshes without folded elements ($\alpha \leq 30°$): smooth field (left three columns) and Lamé (right three columns); PM top, PP bottom.

the interior elements ($q = 2$). The default rule ($n_q = 2p$) is therefore not exact. Repeating selected cases with $n_q = 2p, 3p, 4p, 6p$ (`results/raw/quadrature.csv`; $\alpha = 0°, 30°$, $N = 1, 2$, $p = 5, 8$) changes e_E of the non-polynomial fields by at most $3 \cdot 10^{-7}$ (relative) for PP and 0.5 % for PM; the single exception (PM, smooth field, $\alpha = 30°$, $N = 2$, $p = 8$: 39 %) has an error only six times above the round-off level of that mesh. For the polynomial field (PM, $p \geq 5$), whose error is pure round-off, the value changes by up to a factor 8. The conclusions are therefore not affected by quadrature. On folded meshes the results change erratically with n_q (e.g. polynomial field, $\alpha = 60°$, $N = 2$, $p = 8$: $e_E = 0.78, 0.35, 0.10, 0.38$), consistent with the round-off amplification of Section 4.2, which persists with exact integration.

5 Answers to the key questions

Is distortion resistance lost for non-polynomial fields? Distortion *immunity* in the sense of [5] (results independent of the mesh) is lost by construction: the error of a non-polynomial field depends on the mesh, and it does so for both fields studied (Table 8). Distortion *resistance* depends on the distortion level and on p:

- moderate distortion without folds ($\alpha = 15°$, all $p \leq 6$; $\alpha = 30°$, $p \leq 3$): retained. The energy error of PM exceeds the error on the undistorted mesh by 5–28 % (15°) and 23–84 % (30°, $p \leq 3$), whereas the nodal error of the conventional element is 3–280 times larger ($N = 8$);
- strong distortion ($\alpha = 30°$ with $p \geq 4$, $\alpha \geq 45°$ with $p \geq 3$): lost. The PM error becomes erratic, often larger than that of the conventional element, and on folded meshes ($\alpha \geq 60°$) the PM solutions with $p \geq 5$ are dominated by round-off.

The loss is not tied to the sign of $\det \boldsymbol{J}$, the quantity the formulation was designed to be insensitive to: it already occurs on meshes with $\det \boldsymbol{J} > 0$ ($\alpha = 30°$) and correlates with the Lebesgue constant of the metric interpolation (Table 9).

Are optimal convergence rates retained? In the moderate regime, yes: PM converges with the optimal energy-norm order p for $p = 2, \ldots, 6$ at $\alpha = 15°$ (and $p = 2, 3$ at 30°), for the analytic manufactured field and for the Lamé field, and exponentially under p-refinement. The conventional element loses half of the order on the same meshes ($e_E = O(h^{\lfloor p/2 \rfloor})$), because the element maps are not asymptotically affine. In the strong regime, optimal convergence is lost as well: rates are erratic and the error can grow under h- and p-refinement.

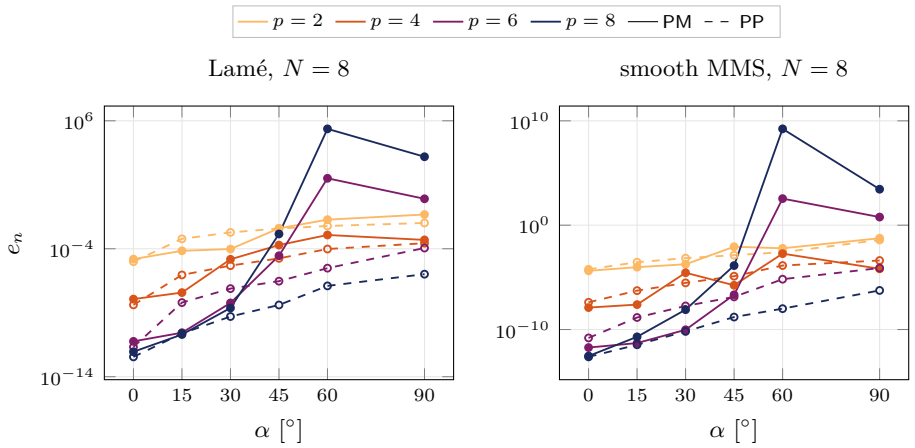


Figure 5: Nodal error versus distortion angle on the finest mesh ($N = 8$), $p = 2, 4, 6, 8$; PM solid, PP dashed. Folded elements occur for $\alpha \geq 45°$.

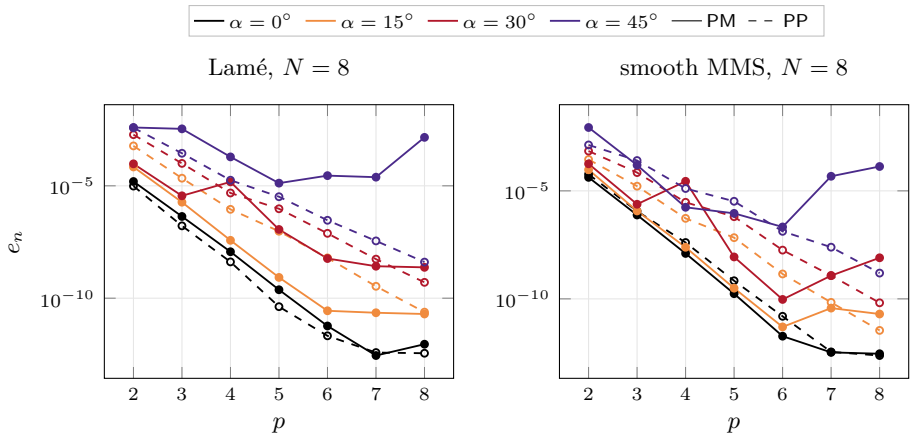


Figure 6: p-convergence of the nodal error on the finest mesh ($N = 8$), $\alpha = 0°, 15°, 30°, 45°$; PM solid, PP dashed.

Practical benefit. Where both properties hold (moderate, non-folding distortion that persists under refinement), the benefit is substantial and grows with p: optimal order versus $\lfloor p/2 \rfloor$, i.e. an error ratio $O(h^{-\lceil p/2 \rceil})$ in favour of PM (a factor 390 for $p = 6$, $N = 8$, $\alpha = 15°$). The benefit is, however, limited in the following respects, all of which concern precisely the situations for which the formulation was proposed:

1. *Jacobian-unlimited meshes are not usable at high order.* For folded meshes the exactness for polynomial fields is destroyed by round-off for $p \geq 5$; for non-polynomial fields neither element is consistently more accurate for $p \leq 4$ (PM has the smaller nodal error in 4 of the 12 cases $p \leq 4$, $\alpha \in \{60°, 90°\}$, $N = 8$), and PM is orders of magnitude less accurate for $p \geq 5$. The conventional spectral element, by contrast, still converges in the nodal error on these meshes for $p \geq 4$.

2. *Lack of robustness.* Whether a given mesh and degree fall into the benign regime cannot be inferred from the Jacobian; non-monotone convergence occurred already at $\alpha = 30°$, $p = 4$. A usable a priori indicator is the Lebesgue constant of the metric interpolation, which is cheap to compute element by element.

3. *Realistic distortion.* The mesh sequence studied keeps the relative distortion fixed under refinement, the worst case for the conventional element. For meshes that become asymptotically affine under refinement (smooth mappings, uniform refinement of a coarse distorted mesh), PP recovers the optimal order, and the advantage of PM reduces to a constant.

4. *Cost.* The unsymmetric system requires a non-symmetric solver (about twice the storage and factorisation cost of a symmetric one), and the metric functions require a generalised Vandermonde solve per element and quadrature point set.

Real-world solution fields are non-polynomial. For them the unsymmetric spectral element is the better choice on moderately distorted, non-folding meshes, where it is distortion tolerant and optimally convergent while

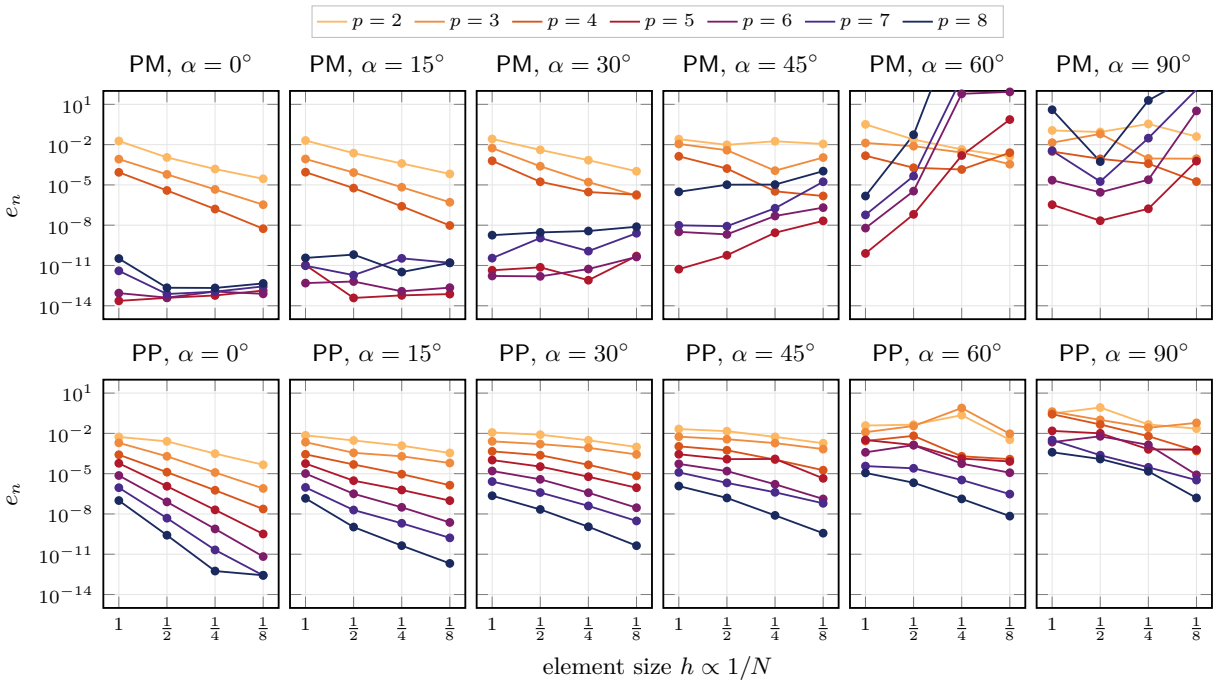


Figure 7: Polynomial field (degree 5): nodal error versus element size, PM (top) and PP (bottom). PM reproduces the field for $p \geq 5$ up to round-off on all meshes without folded elements.

the conventional element is not; it is not a reliable remedy for severely distorted or folded meshes at high polynomial degree, which was the original motivation of the formulation.

6 Limitations

One geometry (thick cylinder), one family of distortions (self-similar rotation of mid-side nodes), linear plane-stress elasticity, $p \leq 8$, $N \leq 8$, double precision, monomial basis of the metric functions (as in `ShapeFunctions2D_Metric.m`); a better-conditioned basis would lower the round-off floor but not the Lebesgue constants, which are basis independent. The energy norm is evaluated with the signed $\det \boldsymbol{J}$; on folded meshes the nodal error is the only quantity that is defined independently of the element mapping. The mechanism of the stability loss (Section 4.5) is characterised numerically, not proven.

7 Reproduction

The results were computed with the code base of commit `2fea864`. Since commit `da84a58`, `ShapeFunctions2D_Metric` stops with an error when its interpolation matrix is singular to working precision (rcond $< \varepsilon$). On the present meshes this concerns PM for $\alpha = 45°$, $p = 8$, $N \geq 4$, and for the folded meshes $\alpha = 60°$ ($p \geq 7, 6, 4, 4$ for $N = 1, 2, 4, 8$) and $\alpha = 90°$ ($p \geq 5, 6, 6, 4$); never for $\alpha \leq 30°$. For $p \geq 5$ the polynomial patch test shows that these results were dominated by round-off (Section 4.2); `run_study.m` records them as NaN when re-run. All other results are unaffected.

`mesh/gen_cylinder_meshes.py`	gmsh model, meshes, ABAQUS files, plot data
`matlab/run_study.m`	main study (MATLAB or GNU Octave)
`matlab/run_interpolation_study.m`	interpolation errors in the trial spaces
`matlab/run_quadrature_study.m`	quadrature sensitivity
`matlab/cyl_verify.m`	cross-checks against the code base
`matlab/cyl_conditioning_check.m`, `cyl_residual_check.m`, `cyl_spectrum_check.m`, `cyl_smin.m`, `cyl_element_analysis.m`	diagnostics of Section 4.5
`postprocess.py`	tables and figure code

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Table 4: Rates of e_E, smooth field ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 0°$	2.00	2.99	4.00	4.99	6.00	–	–
$\alpha = 15°$	2.00	2.99	3.99	4.99	5.98	–	–
$\alpha = 30°$	2.00	2.86	-0.98	2.41	–	–	–
PP (conventional)							
$\alpha = 0°$	2.02	3.19	4.22	5.37	6.39	7.43	–
$\alpha = 15°$	1.08	1.23	2.21	2.03	2.94	3.39	4.05
$\alpha = 30°$	1.02	1.06	2.01	1.98	2.98	3.11	3.94

Table 5: Rates of e_E, Lamé ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 0°$	1.97	2.90	3.89	4.78	5.77	6.62	–
$\alpha = 15°$	1.97	2.91	3.90	4.81	5.83	–	–
$\alpha = 30°$	1.97	2.94	0.76	2.99	4.56	–	–
PP (conventional)							
$\alpha = 0°$	1.91	2.89	3.85	4.79	5.75	6.67	7.60
$\alpha = 15°$	1.07	1.43	2.16	2.81	3.49	4.19	4.78
$\alpha = 30°$	1.03	1.21	2.07	2.51	3.20	3.85	4.42

Table 6: Rates of e_n, smooth field ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 0°$	2.70	3.80	4.86	5.84	6.92	–	–
$\alpha = 15°$	2.86	3.83	4.81	5.79	6.81	–	–
$\alpha = 30°$	2.52	3.30	-1.00	1.97	–	–	–
$\alpha = 45°^\dagger$	2.31	1.12	1.62	-1.00	–	–	–
$\alpha = 60°^\dagger$	-0.37	3.52	-2.08	–	–	–	–
$\alpha = 90°^\dagger$	1.20	-1.23	4.76	–	–	–	–
PP (conventional)							
$\alpha = 0°$	2.91	3.80	4.73	5.78	6.79	–	–
$\alpha = 15°$	1.87	2.41	3.14	2.80	3.66	4.13	4.49
$\alpha = 30°$	1.87	2.00	2.65	2.65	3.61	3.79	4.59
$\alpha = 45°^\dagger$	1.65	1.62	2.28	5.03	3.49	2.84	3.73
$\alpha = 60°^\dagger$	5.79	6.91	0.42	1.39	3.57	3.10	4.01
$\alpha = 90°^\dagger$	-0.26	-2.79	2.45	1.51	3.46	1.62	7.08

Table 7: Rates of e_n, Lamé ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 0°$	3.48	3.98	5.21	5.90	7.22	–	–
$\alpha = 15°$	3.03	3.90	4.96	5.89	6.60	–	–
$\alpha = 30°$	3.19	3.92	0.51	2.84	5.09	–	–
$\alpha = 45°^\dagger$	2.57	1.95	0.87	3.83	-0.28	–	–
$\alpha = 60°^\dagger$	4.35	2.84	5.45	–	–	–	–
$\alpha = 90°^\dagger$	2.23	1.42	5.80	–	–	–	–
PP (conventional)							
$\alpha = 0°$	3.47	4.06	5.31	6.50	7.27	–	–
$\alpha = 15°$	2.02	2.70	2.96	3.15	3.90	4.58	5.13
$\alpha = 30°$	1.88	2.36	3.12	2.91	3.69	4.46	5.12
$\alpha = 45°^\dagger$	1.81	2.17	3.00	2.78	3.59	4.14	4.65
$\alpha = 60°^\dagger$	2.37	-0.59	1.74	2.35	6.75	3.72	1.40
$\alpha = 90°^\dagger$	1.55	-0.48	2.62	2.30	5.29	-0.39	5.21

Table 8: Distortion sensitivity on the finest mesh ($N = 8$): $e_n(\alpha)/e_n(0°)$, smooth field (left) and Lamé (right).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 15°$	2.25	1.54	1.93	1.8	2.68	112	69.1
$\alpha = 30°$	4.24	3.15	2.2e3	50.4	50.2	3.6e3	2.8e4
$\alpha = 45°^\dagger$	204	204	137	5.2e3	1.1e5	1.4e8	4.7e8
$\alpha = 60°^\dagger$	144	636	1.5e5	3.2e9	1.9e14	5.9e18	5.9e21
$\alpha = 90°^\dagger$	1.3e3	2.4e4	5.2e3	5.1e6	3.2e12	6.1e14	9.7e15
PP (conventional)							
$\alpha = 15°$	4.84	14.4	13.2	98.7	91.1	196	14.5
$\alpha = 30°$	11.9	61.2	72.8	923	1.2e3	3.4e3	276
$\alpha = 45°^\dagger$	22.8	221	317	4.8e3	8.6e3	7.0e4	6.6e3
$\alpha = 60°^\dagger$	43.5	7.8e3	3.3e3	4.9e4	4.3e5	1.3e6	4.2e4
$\alpha = 90°^\dagger$	654	8.1e4	1.0e4	3.6e5	5.0e6	3.0e7	2.3e6

	p						
	2	3	4	5	6	7	8
PM (unsymmetric)							
$\alpha = 15°$	4.45	4.31	3.22	3.54	4.74	80	22.1
$\alpha = 30°$	6.11	8.3	1.3e3	481	1.0e3	9.4e3	2.6e3
$\alpha = 45°^\dagger$	253	7.9e3	1.7e4	5.5e4	4.9e6	8.6e7	1.6e9
$\alpha = 60°^\dagger$	1.2e3	1.8e5	1.0e5	2.1e9	5.5e12	2.8e16	2.6e17
$\alpha = 90°^\dagger$	3.0e3	9.1e3	4.2e4	1.4e7	1.4e11	5.4e14	1.7e15
PP (conventional)							
$\alpha = 15°$	61.3	132	218	2.3e3	2.9e3	888	67.5
$\alpha = 30°$	194	601	1.2e3	2.3e4	3.6e4	1.4e4	1.4e3
$\alpha = 45°^\dagger$	392	1.7e3	4.3e3	7.8e4	1.4e5	9.3e4	1.1e4
$\alpha = 60°^\dagger$	619	2.4e4	2.3e4	2.2e5	1.4e6	3.8e5	3.5e5
$\alpha = 90°^\dagger$	1.1e3	2.3e5	6.6e4	2.9e6	5.4e7	9.7e7	2.9e6

Table 9: Lebesgue constant Λ_M of the metric interpolation, maximum over the elements, $N = 8$ (GLL: affine element).

	p						
	2	3	4	5	6	7	8
$\alpha = 0°$	1.56	2.25	2.68	3.16	3.61	4.6	6.05
$\alpha = 15°$	2.54	5.98	15.5	48.2	163	1.4e4	1.3e4
$\alpha = 30°$	10.4	247	1.1e3	1.7e3	1.1e4	4.8e6	2.7e6
$\alpha = 45°^\dagger$	188	1.5e3	2.3e4	8.8e5	5.3e6	4.1e7	4.3e8
$\alpha = 60°^\dagger$	297	1.2e4	1.2e5	1.0e6	1.6e7	2.3e9	5.4e12
$\alpha = 90°^\dagger$	61.5	3.5e3	1.4e6	6.6e5	2.1e6	7.0e8	1.3e9
GLL	1.56	2.25	2.68	3.16	3.46	3.89	4.15

Table 10: Ratio of the maximum element condition numbers $\sigma_{\max}/\sigma_{2n-3}$, PM/PP, $N = 8$.

	p						
	2	3	4	5	6	7	8
$\alpha = 0°$	0.998	1.01	0.996	1	0.997	1.01	0.998
$\alpha = 15°$	1.54	0.937	1.19	1.61	3.25	182	128
$\alpha = 30°$	60	11.5	145	623	90.3	1.4e5	2.1e6
$\alpha = 45°^\dagger$	44.4	12.8	148	7.2e4	1.1e5	6.0e6	2.4e6
$\alpha = 60°^\dagger$	10.4	10.7	104	7.2e3	2.2e5	3.6e7	3.5e10
$\alpha = 90°^\dagger$	9.95	198	23.2	5.7e3	3.7e4	1.6e5	2.8e7